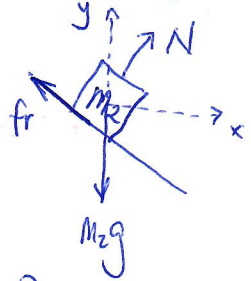


Pauta

Problema 1



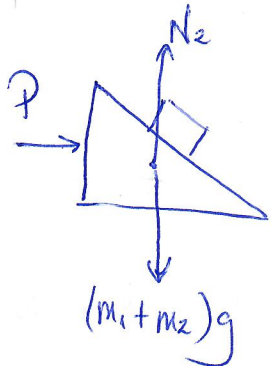
DCL masa m_2

$$\sum F_x \Rightarrow N \sin \theta - f \cos \theta = m_2 a \quad (1)$$

$$\sum F_y \Rightarrow N \cos \theta + f \sin \theta - m_2 g = 0 \quad (2)$$

0.5 cada Σ

Total 2 pto.



DCL sistema $m_1 + m_2$

$$\sum F_x \Rightarrow P = (m_1 + m_2) a \quad (3)$$

$$\sum F_y \Rightarrow N_2 - (m_1 + m_2) g = 0$$

0.5 cada DCL

Total 1 pto

De (2) despejo la Normal N

$$N (\cos \theta + \mu \sin \theta) = m_2 g$$

$$N = \frac{m_2 g}{\cos \theta + \mu \sin \theta} \quad (4)$$

1 pto

Reemplazo (4) en (1)

$$\frac{m_2 g}{\cos \theta + \mu \sin \theta} \cdot \sin \theta - \mu \frac{m_2 g}{\cos \theta + \mu \sin \theta} \cdot \cos \theta = m_2 a$$

$$a = \frac{g}{\cos \theta + \mu \sin \theta} (\sin \theta - \mu \cos \theta)$$

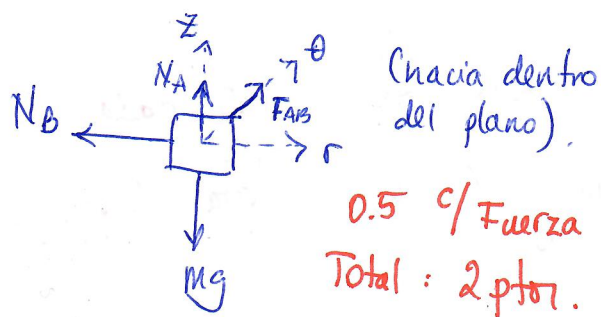
1 pto

Reemplazo a en (3)

$$P = \frac{(m_1 + m_2) g}{\cos \theta + \mu \sin \theta} (\sin \theta - \mu \cos \theta)$$

1 pto.

Problema 2



DCL masa.

La aceleración en coordenadas polares:

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

Como no hay variación en r en el instante en que el anillo está en B

$$a_r = -r\dot{\theta}^2 \quad a_\theta = r\ddot{\theta}$$

$\dot{\theta} = \omega$ y $\ddot{\theta} = \alpha$. Reemplazo lo anterior en ① y ②.

$$N_B = mr\omega^2$$

$$F_{AB} = mr\alpha$$

$$\left. \begin{aligned} \sum F_r &\Rightarrow -N_B = m \cdot a_r \quad (1) \\ \sum F_\theta &\Rightarrow F_{AB} = m \cdot a_\theta \quad (2) \\ \sum F_z &\Rightarrow N_A - mg = 0. \end{aligned} \right\} \begin{aligned} &0.8 \text{ pto} \\ &\text{Total: 2.4 pto} \end{aligned}$$

0.5 por reemplazo
en c/Fuerza